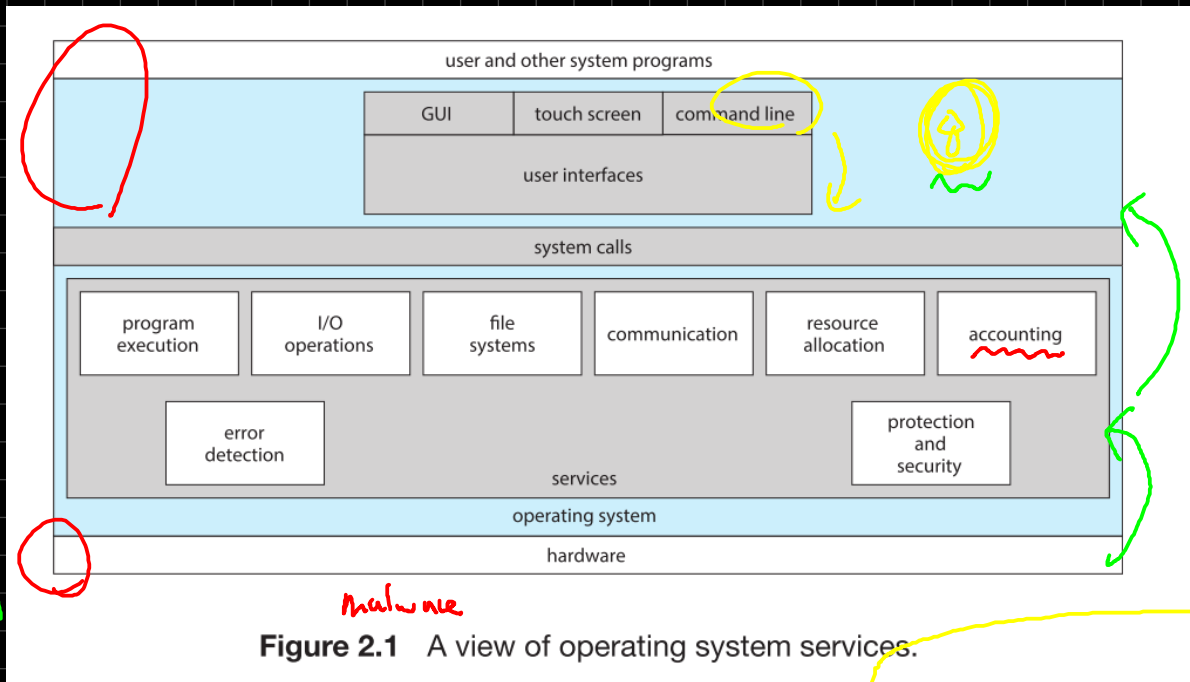




- 4
- (1) between user & hardware
 - (2) control & coordinate resource
 - (3) system view: reliable, easy to maintain, efficient
user view: fast response, easy to use

bug-free

context switch
 process 0 (command line)
 process 1 (mouse event)

Interrupt

PCB (Process Context Block)
 program counter

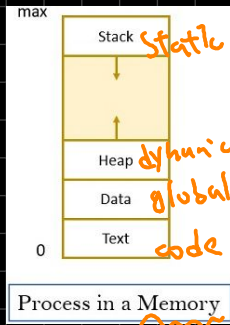
task

process

1. —
 2. —
 3. —
 → 4. —
 5. —

(active) process

DS



static data
dynamic data
global data
code

from disk
load to mem

program

- (1) code (text)
- (2) in disk

* stack / heap 差差?

* thread 有自己?

thread:
lightweight
weight

- (1) shared code, data, heap, open files
- (2) 擁有自己的 program counter, registers, stack

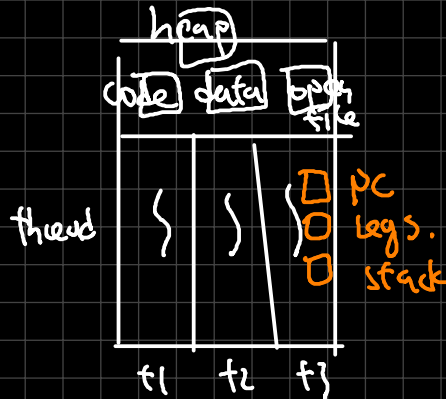
function
local data

thread 是 CPU 執行的基本單位

1 個 process 可以分成好幾個 thread

→ multi-thread programming

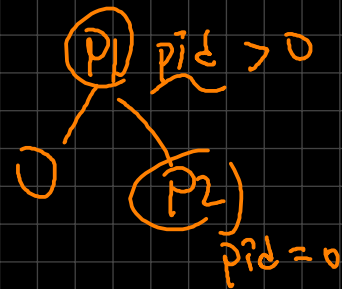
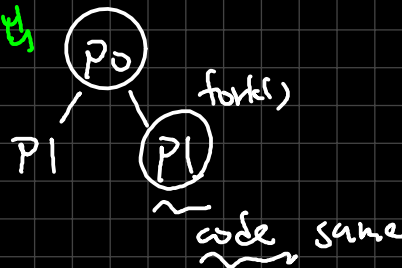
Linux:
thread processing



Unix process creation

- (1) fork() create proc.
pid = 0, child
pid > 0, parent
- (2) execvp() code 替換
- (3) exit()

(1) ctrl + c, 中途退出
(2) OS kill
(這個已經執行完)



* thread 差

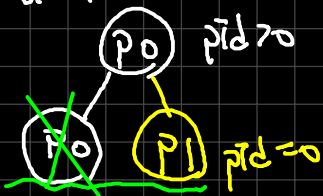
12. Which of the descriptions about the following C program output with printf() are correct?

```
#include <stdio.h>
#include <unistd.h>
void main() {
    pid_t pid;
    int a=2;
    while(a>=0) {
        a--;
        pid = fork();
        printf("pid=%d, a=%d\n", pid, a);
    }
}
```

- ~~(a) Eight lines are output by printf().~~ 14
~~(b) A half of the output "pid" are zero.~~ child (pid=0) $\frac{7}{14} = 0.5$
~~(c) Three processes are created by fork().~~ 7
~~(d) The sum of all output "a" is -6.~~
~~(e) More than one of the processes contains an infinite loop.~~ X

1st

a=1



print: pid=20, a=1
=0 =1

#printed = 2

4th

X

$$\# \text{child} = 1 + 2 + 4 = 7$$

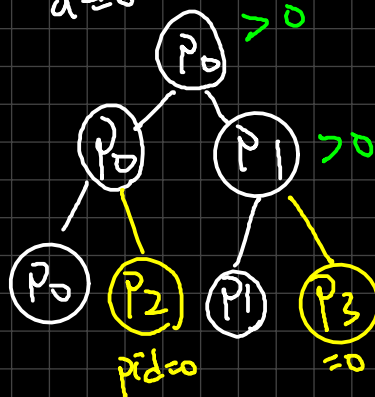
$$\# \text{fork} = 1 + 2 + 4 = 7$$

$$\begin{aligned} \# a &= (1+1) + \\ & (4 \times 0) + \\ & (8 \times -1) \\ &= 2 + 0 - 8 = -6 \end{aligned}$$

fork()

2nd

a=0

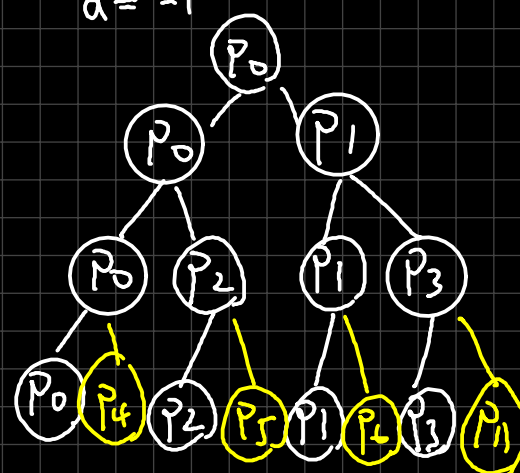


pid=20 = p0, p1
pid=0 = p2, p3

$$\begin{aligned} \# \text{printed} &= 2 + 4 \\ &= 6 \end{aligned}$$

3rd

a=-1



pid=20 = p0, p1, p2, p3
pid=0 = p4, p5, p6, p7

$$\begin{aligned} \# \text{printed} &= 6 + 8 \\ &= 14 \end{aligned}$$

